

Canonical almost-periodic extensions of C -distribution cosine orbits are even

Candidate full answer to a question in arXiv:1808.01197

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Abstract

Kostić, Pilipović, and Velinov ask for conditions under which the unique full-line almost-periodic extension of a positive C -distribution cosine orbit is even. We prove that no extra condition is required: their existing hypotheses already force $\mathbf{C}(-t)x = \mathbf{C}(t)x$. The proof combines recurrent positive almost periods with the cosine identity and closedness of the associated first-order C -distribution-semigroup operator. Passing through the associated 2×2 operator avoids any unsupported assumption that an individual unbounded cosine component is closed.

1 The question and the answer

Let E be a complex Banach space, let $C \in L(E)$ be injective, and let $\mathbf{G}$ be a C -distribution cosine function generated by a closed operator A , with the notation of Kostić–Pilipović–Velinov [1]. Its solution space for the second-order problem is denoted by $Z_2(A)$. For $x \in Z_2(A)$ and $t \geq 0$, write

$$C_x(t) := G(\delta_t)x.$$

The source recalls the invariance and cosine identities

$$G(\delta_t)Z_2(A) \subseteq Z_2(A), \tag{1}$$

$$2G(\delta_s)G(\delta_t)x = G(\delta_{s+t})x + G(\delta_{|s-t|})x \quad (s, t \geq 0). \tag{2}$$

Suppose that $\tilde{E} \subseteq Z_2(A)$ is a linear subspace and that $C_x \in AP([0, \infty); E)$ for every $x \in \tilde{E}$. If

$$F : AP([0, \infty); E) \longrightarrow AP(\mathbb{R}; E)$$

is the source's unique isometric extension map, put $\mathbf{C}_x := F(C_x)$. The question in item 3 on page 10 of [1] asks what conditions ensure $\mathbf{C}_x(-t) = C_x(t)$.

Theorem 1. *Under precisely the hypotheses above, with no additional invariance, closedness, or boundedness assumption on $\tilde{E}$, one has*

$$\mathbf{C}_x(-t) = \mathbf{C}_x(t) = G(\delta_t)x \quad (t \geq 0, x \in \tilde{E}).$$

Thus the canonical almost-periodic extension of every such orbit is even.

3. Here we would like to propose an interesting question with regard to almost periodicity of C -distribution cosine functions. Suppose that $\mathbf{G}$ is a $(C - DCF)$ generated by A , and $\mathbf{G}$ is $\tilde{E}$ -almost periodic. Define $C(t)x := C_x(t) := G(\delta_t)x$, $t \geq 0$, $x \in Z_2(A)$ and $\mathbf{C}(t)x := [F(C_x(\cdot))](t)$, $t \in \mathbb{R}$, $x \in \tilde{E}$. As in the semigroup case, we have that $\mathbf{C}(\cdot)$ commutes with C as well as that $\|\mathbf{C}(t)x\| \leq \sup_{s \geq 0} \|C(s)x\|$, $x \in \tilde{E}$, $t < 0$, and $\sup_{t \in \mathbb{R}} \|\mathbf{C}(t)x\| = \sup_{t \geq 0} \|C(t)x\|$, $x \in \tilde{E}$. In the present situation, we do not know to tell what conditions ensure the validity of expected equality $\mathbf{C}(-t)x = C(t)x$, for $t \geq 0$ and $x \in \tilde{E}$.

Figure 1: The exact question in [1, p. 10, item 3].

2 Two elementary inputs

We isolate the two points that make the limit argument rigorous.

Lemma 1 (half-line periods remain full-line periods). *Let $f \in AP([0, \infty); E)$ and $g = Ff$. For every $\tau \geq 0$,*

$$\sup_{r \in \mathbb{R}} \|g(r + \tau) - g(r)\| = \sup_{r \geq 0} \|f(r + \tau) - f(r)\|.$$

Proof. The full-line almost-periodic function $r \mapsto g(r + \tau) - g(r)$ restricts on $[0, \infty)$ to $r \mapsto f(r + \tau) - f(r)$. By uniqueness of F , it is the F -extension of that restriction. The equality is exactly the isometry of F . $\square$

For the second input, recall the source's first-order reduction

$$\mathcal{A} = \begin{pmatrix} 0 & I \\ A & 0 \end{pmatrix}, \quad \mathcal{C} = \begin{pmatrix} C & 0 \\ 0 & C \end{pmatrix}.$$

The associated $\mathcal{C}$ -distribution semigroup on $E \oplus E$ will be denoted by $\mathcal{G}$. For every compactly supported distribution T , its operator $\mathcal{G}(T)$ is a closed linear operator by the general construction recalled in [1, Section 1].

Lemma 2 (orbit of the first-order reduction). *For $y \in Z_2(A)$ and $t \geq 0$,*

$$\mathcal{G}(\delta_t) \begin{pmatrix} 0 \\ y \end{pmatrix} = \begin{pmatrix} \int_0^t G(\delta_s)y \, ds \\ G(\delta_t)y \end{pmatrix}. \quad (2.1)$$

Proof. Put $u(s) = G(\delta_s)y$ and $U(s) = (\int_0^s u(r) \, dr, u(s))^T$. The second-order mild identity gives

$$\int_0^t (t - r)u(r) \, dr \in D(A), \quad A \int_0^t (t - r)u(r) \, dr = u(t) - y.$$

Consequently

$$\mathcal{A} \int_0^t U(s) \, ds = \begin{pmatrix} \int_0^t u(s) \, ds \\ u(t) - y \end{pmatrix} = U(t) - U(0).$$

Thus U is the first-order mild solution with initial value $(0, y)^T$. The source characterization of the solution space of a distribution semigroup identifies it with the orbit of $\mathcal{G}(\delta_t)$, proving (2.1). $\square$

3 Proof of Theorem 1

Fix $x \in \tilde{E}$ and abbreviate

$$f(s) = G(\delta_s)x \quad (s \geq 0), \quad g = Ff.$$

Fix $t > 0$. For each n , relative density of the positive $1/n$ -periods of f lets us choose such a period $\tau_n > t + n$. By Lemma 1,

$$\sup_{r \in \mathbb{R}} \|g(r + \tau_n) - g(r)\| \leq \frac{1}{n}. \quad (3)$$

In particular, with $y_n = f(\tau_n)$, we have $y_n \rightarrow g(0) = x$.

By (1), $y_n \in Z_2(A)$. For $0 \leq s \leq t$, the choice $\tau_n > t$ and (2) give

$$\begin{aligned} G(\delta_s)y_n &= \frac{1}{2}(f(\tau_n + s) + f(\tau_n - s)) \\ &= \frac{1}{2}(g(\tau_n + s) + g(\tau_n - s)). \end{aligned}$$

The full-line estimate (3) therefore yields, uniformly for $0 \leq s \leq t$,

$$G(\delta_s)y_n \longrightarrow p(s) := \frac{1}{2}(g(s) + g(-s)). \quad (4)$$

Apply Lemma 2. Uniform convergence in (4) shows that

$$\mathcal{G}(\delta_t) \begin{pmatrix} 0 \\ y_n \end{pmatrix} = \begin{pmatrix} \int_0^t G(\delta_s)y_n ds \\ G(\delta_t)y_n \end{pmatrix} \longrightarrow \begin{pmatrix} \int_0^t p(s) ds \\ p(t) \end{pmatrix}.$$

Since $(0, y_n)^T \rightarrow (0, x)^T$ and the operator $\mathcal{G}(\delta_t)$ is closed, the limit is $\mathcal{G}(\delta_t)(0, x)^T$. Lemma 2, now for x , identifies its second coordinate as

$$p(t) = G(\delta_t)x = f(t) = g(t).$$

By the definition of $p(t)$ this is $\frac{1}{2}(g(t) + g(-t)) = g(t)$, hence $g(-t) = g(t)$. The case $t = 0$ is immediate, completing the proof. $\square$

4 Checks and consequences

- The proof never applies a closed-graph argument directly to the possibly unbounded component $G(\delta_t)$. It proves convergence of both coordinates and uses the explicitly closed associated operator $\mathcal{G}(\delta_t)$.
- No assumption that $G(\delta_t)\tilde{E} \subseteq \tilde{E}$ is needed. Only the source's automatic invariance of $Z_2(A)$ is used.
- The argument is orbitwise. It does not require $\tilde{E}$ to be closed or dense and does not require uniform operator bounds.
- The theorem answers the displayed even-extension question only. It does not address the distinct weak-almost-periodicity problem at the end of the source paper.

References

- [1] M. Kostić, S. Pilipović, and D. Velinov, *Subspace Almost Periodic C -Distribution Semigroups and C -Distribution Cosine Functions*, Applied Mathematics and Computer Science **2** (2017), no. 1, 1–12; [arXiv:1808.01197](#).